MILITARY SPECIFICATION

TRANSPONDER, RADAR TRACKING

1. SCOPE

1.1 Scope. This specification establishes the requirements for the Radar Tracking Transponder, hereinafter called the transponder (see 6.1).

1.2 Classification. The transponders covered by this specification are classified as follows:

a. Type I - 5.4 to 5.9 Gigahertz (GHz)

b. Type II - 9.0 to 9.5 GHz

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-W-5088 Wiring, Aerospace Vehicle

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, New Jersey 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-E-5400	Electronic Equipment, Aerospace, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft, Electronic Equipment
MIL-E-17555	Electronic and Electrical Equipment, Accessories and Repair Parts; Packaging and Packing of
MIL-T-18303	Test Procedures: Preproduction, Acceptance and Life for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Identification for Electronics, Aeronautical and Aeronautical Support Equipment Including Ground Support Equipment

STANDARDS

MILITARY

MIL-STD-108	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	Aircraft Electric Power Requirements
MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution
MIL-STD-785	Reliability Program for System Equipment Development and Production
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-2074	Failure Classification for Reliability Testing

FEDERAL

FED-STD-595	Colors

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2.1.2 Other Government documents, drawings, and publications. The following drawings form a part of this specification to the extent specified herein.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND
(Code Ident. 30003)

1356AS3020 Transponder, Radar Tracking

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Parts and materials. In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

a. Microelectronic technology shall be considered and microelectronic items shall be in accordance with requirements specified herein.

b. Other parts and materials requirements shall be in accordance with MIL-E-5400.

c. Nonrepairable subassemblies shall be used in accordance with MIL-E-5400.

d. Parts shall be selected using the following order of preference:

(1) Parts listed on a qualified parts list (QPL) with two or more sources.

(2) Parts listed on a QPL with only one source.

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(3) Commercial parts with two or more sources.

(4) Commercial parts with only one source.

3.2.1 Standard parts. Parts specified in 3.2(d)(1) are standard parts and do not require approval from the procuring activity. Parts selected for use from 3.2(d)(2), (3) and (4) are considered nonstandard and require approval. Nonstandard parts approval requests shall be prepared using DD Form 2052 in accordance with the contract (see 6.2.2).

3.2.2 Commercial parts. Commercial parts specified in 3.2(d)(3) and (4) shall be readily available, nationally catalogued, off the shelf items.

3.2.3 Development contractor parts. Parts manufactured by the development contractor only will not be accepted without full design disclosure including any special processing required to duplicate the part.

3.2.4 Proprietary parts. Proprietary parts, assemblies or processes shall not be used.

3.2.5 Nonstandard parts. Use of nonstandard parts in first article or production units prior to Government approval is at the sole risk of the contractor and subject to replacement if the government does not approve proposed nonstandard part.

3.2.6. Modules. Conformal coating, encapsulants, embedments or potting materials used with modular assemblies containing integrated circuits and discrete parts shall be easily removable without damage to the assembly. The transponder shall be logically modularized for replacement of sub-assembly and ease of trouble shooting for maintenance. All printed circuit boards shall be mounted with circuit side facing toward the top, with accessibility of test points provided with the cards in place.

3.3 Design and construction. The transponder shall be in accordance with all the applicable requirements of MIL-E-5400 for design and construction, except as otherwise specified herein.

3.3.1 Total weight. The total weight of the transponder shall be a minimum consistent with good design and shall be not greater than 1275 grams (gm) [45 ounces (oz)].

3.3.2 Reliability. The contractor shall conduct a reliability program in accordance with MIL-STD-785. On a reorder from a contractor who is producing the transponder, the program previously approved (see 6.2.2) may be continued unless otherwise specified in the contract or purchase order (see 6.2.1d).

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3.3.2.1 Operational stability. The transponder shall operate as specified herein, continuously or intermittently for a period of not less than 25 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use. Note: The tuning ports shall be considered inaccessible during normal use for the requirements of this paragraph.

3.3.2.2 Operating life. The transponder shall have a total operating life of not less than 1000 hours with minimal servicing and replacement of parts. The contractor shall prepare a repairable parts list for review by the procuring activity (see 6.2.2). This list shall identify parts requiring scheduled replacement.

3.3.2.3 Reliability in mean time between failure (MTBF). The transponder, including any built-in-test provisions, shall have a lower test MTBF of 50 hours and an upper test MTBF of 100 hours when tested as specified in 4.5.2.

3.3.3 Cabling and connections. Cabling and connections shall be as specified in 3.3.3.1 and 3.3.3.2.

3.3.3.1 Cables and connectors. The transponder shall provide for the use of wiring and connectors in accordance with MIL-E-5400.

3.3.3.2 Interconnection cable. The transponder shall operate as specified herein, using external wiring in accordance with the requirements of MIL-W-5088.

3.3.4 Interchangeability. The transponder shall be in accordance with the interchangeability requirements of MIL-E-5400.

3.3.5 Interference control. The generation of radio interference by the transponder and vulnerability of the transponder to radio interference shall be in accordance with the requirements of MIL-STD-461, except as specified herein.

3.3.6 Provisions for maintainability. The contractor shall provide for maintainability in accordance with the requirements of MIL-E-5400.

3.3.7 Nomenclature, nameplates, and identification markings. Nomenclature and serial number assignment shall be in accordance with MIL-N-18307 (see 6.2.2 and 6.4). Nameplate shall be in accordance with Drawing 1356AS3020.

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3.3.8 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

a.	Temperature:	Room ambient (25°C ± 10°C)
b.	Altitude:	Normal ground
c.	Vibration:	None
d.	Humidity:	Room ambient up to 95 percent relative humidity
e.	Input voltage:	+28.0 ± 0.5 volts direct current (Vdc)

3.3.9 Service conditions. The transponder shall operate as specified herein, under any of the environmental service conditions or reasonable combination of these conditions in accordance with MIL-E-5400 for Class 2 equipment, except as specified herein.

3.3.9.1 Vibration. The transponder shall operate as specified herein when subjected to the vibration requirements of Figure 2, curve IVa of MIL-E-5400. The limit shall be ±10 gravity units (g's).

3.3.9.2 Shock. The transponder shall operate as specified herein when subjected to shock conditions with peak amplitudes of 100 g's for 11 milliseconds (msecs) in any axis.

3.3.9.3 Submergence. The transponder shall withstand submergence in three feet of water for one hour without water intrusion and shall operate as specified herein after removal. The transponder's exterior surfaces shall show no signs of corrosion after immersion in sea water.

3.3.9.4 Temperature and temperature - altitude. Intermittent operation at +95°C is not required.

3.3.9.5 Altitude. The transponder shall operate as specified herein at any altitude up to and including 100,000 feet.

3.3.10 Warm-up time. The time required for the transponder to warm-up prior to operation shall be kept to a minimum and shall be not greater than three minutes under either standard or service conditions.

3.3.11 Input electrical power. The electrical power shall be as specified in 3.3.11.1 through 3.3.11.3.

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3.3.11.1 Operating power. The transponder shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following power sources having characteristics and limits as defined in MIL-STD-704, except as specified herein. The power required shall be not greater than the following amounts:

a. 28 Volts normal dc power ranging from 21 to 31 Vdc.

b. 1.0 amperes maximum current at quiescent.

c. 1.5 amperes current maximum at pulse repetition frequency of 3000 Hertz (Hz).

3.3.11.2 Degraded performance. Degraded performance is permissible for all voltage transients from 50 microseconds (μsec) to 0.5 seconds duration in accordance with MIL-STD-704 for normal electric system operation. Operation shall return to normal, with no resulting damage to the transponder, within 0.5 seconds after the 50 μsec to 0.5 second transient expires.

3.3.11.3 Voltage protection. Reverse polarity protection on the +28 Vdc input shall be provided.

3.3.12 Form factor. The transponder shall be in accordance with Drawing 1356AS3020.

3.3.13 Pressurization. The transponder shall be capable of being pressurized to 16.0 ± 0.5 pounds per square inch (psi).

3.3.13.1 Pressure differential. The case shall withstand a positive or negative five psi pressure difference over the applicable pressure range.

3.3.13.2 Case type. The case shall be of a type that shall permit ready opening and clearing for access to the equipment for repair and maintenance.

3.3.13.3 Effectiveness of the seal. A means shall be provided on the transponder to determine the effectiveness of the seal. This may consist of an automobile-tire-type valve stem fitting to permit the use of an air pump for increasing the pressure approximately five psi above sea level pressure. Measure of the pressure shall be by means of a Schrader type 3715 gage or equivalent.

3.3.13.4. Maintenance of pressure. The transponder shall maintain operating pressure for periods of not less than 24 hours under any service condition.

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3.4 Performance. Unless otherwise specified herein, values specified to establish required performance apply to performance under both standard and extreme environmental conditions.

3.4.1 Operation. The transponder shall not be provided with a power switch and is to be activated when power is applied to the host vehicle.

3.4.2 Frequency characteristics. The transponder shall be capable of receiving and transmitting within the frequency specified for each respective type:

a. Type I - 5.4 to 5.9 GHz

b. Type II - Receive 9.0 to 9.5 GHz and transmit 9.2 to 9.5 GHz.

Separation between transmit and receive frequencies shall be not less than 50 Megahertz (MHz).

3.5 Detail requirements. Detail requirements shall be as specified in 3.5.1 through 3.5.13.

3.5.1 Contents. The transponder shall contain the circuit groups specified in 3.5.1.1 through 3.5.1.3.

3.5.1.1 Receiver. The receiver shall be of the superheterodyne type and capable of being tuned to any frequency from 5.4 to 5.9 GHz for Type I, and 9.0 to 9.5 GHz for Type II.

3.5.1.1.1 Factory tuning. Unless otherwise specified in the contract or purchase order, the transponder will have the receiver tuned to 5505 MHz ± 100 KHz for Type I and 9225 MHz ± 100 KHz for Type II (see 6.2.1e).

3.5.1.1.2 Receiver characteristics. The receiver characteristics shall be as specified in 3.5.1.1.2.1 through 3.5.1.1.2.5.

3.5.1.1.2.1 Interrogation signal. The transponder shall receive radar interrogation signals. The interrogation signal may be either single or paired pulses as specified in 3.5.1.2. The choice of the single pulse signal shall be accomplished by internal soldering of a jumper wire.

3.5.1.1.2.2 Sensitivity. When an interrogation signal at a level between -70 for Type I and -65 for Type II decibels below one milliwatt (dBm) and 0 dBm is applied to the transponder antenna connector, the transponder shall trigger at least 99 percent replies.

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3.5.1.1.2.3 Frequency stability. The center frequency shall not vary more than ±2 MHz when the transponder is operated in accordance with 3.3.2.1.

3.5.1.1.2.4 Bandwidth. The transponder shall have a bandwidth of 11 ±3 MHz at the 3 decibel (dB) points.

3.5.1.1.2.5 Image frequency rejection. The rejection of image frequencies need not be greater than 60 dB.

3.5.1.2 Decoder characteristics. The decoder characteristics shall be in accordance with 3.5.1.2.1 through 3.5.1.2.3.

3.5.1.2.1 Single pulse interrogation. When the transponder is modified for this mode of operation, the transponder shall respond to a single pulse interrogation having the following characteristics:

a. Pulse width: 1.0 ±0.2 µsec

b. Pulse rise time: not greater than 0.1 µsec

3.5.1.2.2 Double pulse interrogation. The transponder shall respond to a double pulse interrogation having the following characteristics:

a. Pulse space: shall be adjustable from 3.0 to 12.0 µsec

b. Pulse width: 0.25 to 0.5 µsec

c. Pulse rise time: not greater than 0.1 µsec

3.5.1.2.2.1 Double pulse reject limit. The transponder shall be capable of rejecting interrogations having a spacing of ±0.3 µsec of the chosen spacing.

3.5.1.2.2.2 Factory tuning of pulse spacing. Unless otherwise specified in the contract or purchase order, the transponder will have the pulse spacing set to 7.0 ±0.1 µsec for Type I, and 4.8 ±0.1 µsec for Type II (see 6.2.1f).

3.5.1.2.3 Random triggering. The tranponder shall exhibit a random triggering of not greater than 10 per minute averaged over a 10 minute interval.

3.5.1.3 Transmitter characteristics. The transmitter characteristics shall be as specified in 3.5.1.3.1 through 3.5.1.3.6.

3.5.1.3.1 Transmitter. The transmitter shall be of a magnetron type capable of being tuned to any frequency from 5.4 to 5.9 GHz for Type I and 9.2 to 9.5 for Type II.

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3.5.1.3.1.1. Factory tuning. Unless otherwise specified in the contract or purchase order, the transponder will have the transmitter tuned to 5650 MHz ± 100 KHz for Type I and 9325 MHz ± 100 KHz for Type II (see 6.2.1g).

3.5.1.3.2 Frequency stability. The transmitter frequency shall be within ±2.5 MHz of the initial setting under all conditions.

3.5.1.3.3 Transmitter power. The peak pulse power from the transmitter antenna terminal shall be not less than 400 watts at 500 pulse repetition frequency (prf) into a load with a voltage standing wave ratio (VSWR) of 1.5:1.

3.5.1.3.4 Reply signal characteristics. The transponder shall reply to properly coded interrogations with a single pulse having the following characteristics:

a. Pulse width: 0.5 ±0.1 µsec

b. Pulse rise time (10 to 90 percent): not greater than 0.1 µsec

c. Pulse fall time (90 to 10 percent): not greater than 0.20 µsec

d. Leading edge jitter: ± 0.05 µsec at an input signal level of -55 dBm to -65 dBm.

± 0.02 µsec at an input signal level of -55 dBm to 0 dBm.

3.5.1.3.5 Over interrogation protection. The transponder shall not suffer permanent damage when subjected to interrogations which would result in exceeding the duty cycle limitation of 0.002. This circuitry may utilize the recovery time of the transponder provided there is no degradation in performance from that specified in 3.4.

3.5.1.3.6 Pulse repetition frequency. The transmitter shall be capable of operating at a rate of up to 3000 pulses per second.

3.5.2 Transponder recovery time. The transponder recovery time shall be not greater than 50 µsec for input signal levels differing by up to 65 dB.

3.5.3 Transponder delay time. The fixed delay time between the single interrogation pulse or the second pulse of the double pulse interrogation and the beacon reply pulse shall be adjustable within the limits of 1.5 to 6.0 µsec.

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3.5.3.1 Delay variation with signal level. The total variation in delay shall be not greater than ±0.05 µsec for signal levels of 0 dBm to -65 dBm.

3.5.3.2 Factory tuning. Unless otherwise specified in the contract or purchase order, the transponder delay time shall be 2.0 ±0.01 µsec at 0dBm for Type I and Type II (see 6.2.1h).

3.5.4 Random triggering. The transponder shall not trigger due to random noise when the receiver radio frequency (RF) input is terminated with a dummy load.

3.5.5 Antenna terminal. The receiver and transmitter shall have one antenna connection.

3.5.6 Maximum signal input. The transponder shall not be damaged by input signal levels of not greater than +20 dBm.

3.5.7 Antenna system compatibility. The transponder shall operate as specified in conjunction with an antenna system having a VSWR of not less than 2:1.

3.5.8 Impedance. The input and output impedance of the transponder shall be 50 ohms.

3.5.9 Controls. The controls shall be as specified in 3.5.9.1 through 3.5.9.2.

3.5.9.1 Internal tuning adjustments. The following controls shall be provided internal to the transponder. All adjustments shall be accessible with the top cover removed:

a. Receiver frequency control

b. Transmitter frequency control

c. Decoder pulse spacing control

d. Transponder delay time control

3.5.9.2. External tuning adjustments. In addition to meeting the requirements of 3.5.9.1, the following controls shall be accessible through tuning ports in the cover (same cover as defined in 3.5.9.1).

a. Decoder pulse spacing control

b. Transponder delay time control

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3.5.10. Electrical connections. The transponder shall be equipped with the connectors specified in Table I. Pin assignments for J2 are specified in Table II.

Table I. Electrical connections.

Reference designation	Type	Function
J1	Type TNC female	Antenna port
J2	MS3116E8-4P	Power

Table II. J2 pin assignments.

Pin	Function
A	+28VDC
B	ground 1/
C	spare
D	ground 1/

1/ Pins B and D shall be internally connected.

3.5.11 Finish and color. The transponder finish and color shall be in accordance with FED-STD-595 as follows:

a. Type I, semi-gloss, blue 25053

b. Type II, semi-gloss, grey 26307

c. Lettering Type I, gloss, white 17886

d. Lettering Type II, gloss, black 17038

3.5.12 Soldering. Soldering shall be in accordance with MIL-STD-454, Requirement 5.

3.5.13 Workmanship. The transponder shall be manufactured and assembled in accordance with MIL-STD-454, Requirement 9.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1i), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order (see 6.2.1), the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in 3.3.9 and 4.5.2. Inspections to demonstrate compliance with the requirements of MIL-E-5400 shall be in accordance with procedures of MIL-T-5422, except three cycles of the humidity tests shall be performed instead of 10.

4.4 First article inspection. When specified in the contract or purchase order (see 6.2.1j), first article inspections shall be conducted by the contractor on a transponder representative of the production transponders to be supplied under the contract. The first article inspection shall consist of the tests specified in Table III and shall be performed in the sequence listed.

Table III. First article inspection.

Test Requirement	Requirement paragraph	Test method paragraph
1. Product Examination		4.4.3.1
a. Physical characteristics	3.2, 3.3.1, 3.3.12	
b. Name plate	3.3.7	
c. Connectors	3.5.10	
d. Finish and color	3.5.11	

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Table III. First article inspection. (continued)

Test Requirement	Requirement paragraph	Test method paragraph
2. Performance		4.4.3.2
a. Operational stability	3.3.2.1	
b. Warm-up time	3.3.10	
c. Input power	3.3.11	
d. Performance	3.4	
e. Detail requirements	3.5	
3. Electromagnetic interference control	3.3.5	4.4.3.4
4. Environmental		4.4.3.3
a. Service conditions	3.3.9	
b. Warm-up	3.3.10	
5. Submergence	3.3.9.3	4.4.3.5
6. Air tightness	3.3.13	4.4.3.6
7. Reliability	3.3.2.3	4.5.2

4.4.1 First article inspection data. The contractor shall prepare a first article inspection report including all data collected in conducting these inspections (see 6.2.2).

4.4.2 First article approval. Approval of the first article will be by the procuring activity upon satisfactory completion of all inspections. No production transponders shall be delivered prior to the approval of the first article. Prefabrication of production transponders prior to the approval of the first article is at the contractor's own risk. The approved first article shall be retained by the contractor for his use in the fabrication and testing of transponders to be submitted for acceptance. The first article shall not be considered as one of the production transponders under the contract.

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4.4.3 First article test method. First article test methods shall be as specified in 4.4.3.1 through 4.4.3.6.

4.4.3.1 Examination of product. The contractor shall examine the transponder to ensure that all product examination requirements specified in Table III have been met.

4.4.3.2 Performance. The contractor shall conduct this test to demonstrate that all performance requirements specified in Table III have been met at standard conditions.

4.4.3.3 Environmental. The contractor shall conduct tests to demonstrate compliance with the requirements of MIL-E-5400, using the procedures of MIL-T-5422, except three cycles of the humidity tests shall be performed instead of 10.

4.4.3.4 Electromagnetic interference. The contractor shall conduct tests to demonstrate compliance with the requirements of MIL-STD-461 have been met. Tests shall be performed in accordance with MIL-STD-462.

4.4.3.5 Submergence. The contractor shall conduct tests to demonstrate compliance with the submergence requirement of Table III. Tests shall be performed in accordance with MIL-STD-108.

4.4.3.6 Airtightness. The contractor shall conduct tests to demonstrate compliance with the airtightness requirement of Table III. Tests shall be performed in accordance with MIL-STD-108.

4.5 Quality conformance inspections. The contractor shall furnish all samples and shall be responsible for accomplishing the quality conformance inspections. All inspection and testing shall be subject to monitoring by the Government inspector. The contractor shall prepare test reports showing quantitative results for all quality conformance inspections (see 6.2.2). Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Quality conformance inspections shall consist of the following:

a. Individual tests (see 4.5.1)

b. Reliability assurance tests (see 4.5.2)

c. Special tests (see 4.5.3)

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4.5.1 Individual tests. Each transponder submitted for acceptance shall be subjected to the individual tests. These tests shall demonstrate compliance with the requirements of material, operational performance and reliability. As a minimum, each transponder accepted shall have passed the following tests:

a. Examination of product (see 4.5.1.1)

b. Operational test (see 4.5.1.2)

c. All transponder production reliability acceptance test (see 4.5.1.3).

d. Seal verification test (see 4.5.1.4)

4.5.1.1 Examination of product. Each transponder shall be examined carefully to determine that the material and workmanship requirements have been met.

4.5.1.2 Operational test. The contractor shall conduct this test on each transponder offered for acceptance in accordance with performance requirements specified in Table IV. These tests are to be conducted under standard conditions.

Table IV. Operational test.

Test requirement	Requirement paragraph
1. Warm-up time	3.3.10
2. Operating power	3.3.11.1
3. Receiver frequency	3.5.1.1.1
4. Receiver sensitivity	3.5.1.1.2.2
5. Double pulse interrogation	3.5.1.2.2
6. Random triggering	3.5.1.2.3
7. Transmitter frequency	3.5.1.3.1.1
8. Transmitter power	3.5.1.3.3
9. Reply signal characteristics	3.5.1.3.4
10. Delay variation with signal level	3.5.3.1
11. Delay time setting	3.5.3.2

4.5.1.3 All transponder production reliability acceptance test. Each transponder produced, except those submitted for the qualification phase (see 4.5.2.1) shall be tested for 25 failure free hours of operating time.

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The test level shall be the same as the qualification phase. Prior to the 25 hours failure free test, a burn-in period may be used. If the burn-in period is used, the details thereof shall be included in the procedures. To determine whether the MTBF is being met at any time during the contract, the operating test hours and the failures thereon (not counting burn-in failures or burn-in operating time) shall be totaled and the results compared with the accept-reject criteria of Figure C-13 of MIL-STD-781. These totals shall accummulate so that at any one time the experience from the beginning of the contract is included. The procuring activity shall be notified any time that the current totals of test hours and test failures plotted on Figure C-13 of MIL-STD-781 shows a reject situation. The procuring activity reserves the right to stop the acceptance of transponders at any time that a reject situation exists pending a review of the contractor's effort to improve the transponder, the transponder parts, the transponder workmanship, so that the entire compilation will show other than a reject decision.

4.5.1.4 Seal verification test. Each transponder shall be pressurized to 16 +0.5 psi and shall be completely immersed in water for a period of not less than one minute. The transponder shall be observed for visible signs of leakage (bubbles). Any visible signs of leakage (bubbles) shall be considered a failure.

4.5.2 Reliability assurance tests. Reliability assurance tests shall be conducted in accordance with MIL-STD-781 unless otherwise specified in the contract or purchase order (see 6.2.1k), tests as required by the qualification phase shall be conducted. Classification of failures shall be in accordance with MIL-STD-781 and MIL-STD-2074 (see 3.3.2.3). The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, and preventive maintenance to be allowed during the test, shall be part of the test procedures to be prepared prior to the beginning of the reliability assurance tests.

4.5.2.1 Qualification phase. Prior to the acceptance of transponders under the contract or purchase order, no less than three transponders shall be tested and shall have passed in accordance with test XIVC (Chart 15) of MIL-STD-781. For the qualification phase, the test level shall be:

a.	Temperature (cycling):	-40 to +55°C
b.	Vibration:	2.2 g ±10 percent at any non-resonant frequency between 20 and 60 Hz
c.	Input voltage:	+27.0 to +29.5 Vdc

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d. Transponder on-off cycling: Transponder off during cooling cycle and on during heating cycle

4.5.2.2 Test details. The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, and preventive maintenance to be allowed during the test, shall be part of the test procedures to be prepared prior to the beginning of the qualification test phase of the reliability assurance tests (see 6.2.2).

4.5.3 Special tests. Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the transponder and to assure adequate quality control. The transponder selected for special tests may be selected from transponders previously subjected to the sampling or reliability assurance tests.

4.5.3.1 Special test schedule. Selection of transponders for special tests shall be made as follows:

a. On one of the first 10 transponders after an engineering or material change.

b. Whenever failure reports or other information indicates that additional tests are required. (This will be determined by the procuring activity.)

4.5.3.2 Scope of test. Special tests shall consist of such tests as required by the procuring activity. Test procedures previously approved shall be used where applicable. When not applicable, the contractor shall prepare a test procedure prior to conducting the tests.

4.5.4 Transponder failure. Should a failure occur during either the reliability assurance or special tests, the following action shall be taken:

a. Determine the cause of failure.

b. Determine if the failure is due to a material defect or a design defect.

c. Prepare for procuring activity acceptance, the proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests (see 6.2.2).

d. Where practical, include a test in the individual test to check all transponders for this requirement until assurance is obtained that the defect has been corrected.

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4.6 Test procedures. The procedures for conducting first article and quality conformance inspections shall be prepared by the contractor (see 6.2.2). The right is reserved by the procuring activity to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided by the procuring activity and may be used. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary (see 6.2.1m).

4.7 Production transponders. Transponders supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equal to the approved first article. Each transponder shall be capable of successfully passing the same tests as imposed on the first article. Evidence of noncompliance with the above shall constitute cause for rejection, and, for transponders already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections.

4.8 Presubmission testing. No item, part, or complete transponder shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply with the applicable requirements of this specification and the contract.

5. PACKAGING

5.1 General. The transponder shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or purchase order (see 6.2.1n) in accordance with MIL-E-17555 and MIL-STD-794. The method of preservation for level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794. Plastic protective caps shall be provided on all connectors.

6. NOTES

6.1 Intended use. The transponders covered by this specification are intended to receive interrogations from radars and transmit a reply to the interrogation on another frequency. The transponders are intended to be used to extend the operating range of the radar.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Number of transponders to be procured.

c. Type I or Type II (see 1.2).

d. Prohibition of the use of previously approved reliability program (see 3.3.2), if applicable.

e. Specific receiver tuning frequency, if other than specified in 3.5.1.1.1.

f. Specific width of pulse spacing, if other than specified in 3.5.1.2.2.2.

g. Specific transmitter tuning freqency, if other than specified in 3.5.1.3.1.1.

h. Delay time if other than specified in 3.5.3.2.

i. Responsibility for performance and facilities for inspection if other than specified in 4.1.

j. First article inspection, if required (see 4.4).

k. Reliablity assurance test if other than specified in 4.5.2.

m. Preparation of test procedures if other than specified in 4.6 and additional tests, if required.

n. Level of preservation-packaging (see 5.1).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR-7-104.9 (n)(2) are invoked and the DD Form 1423 is not used, the data

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specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph No.	Data requirement title	Applicable DID No.	Option
3.2.1	Nonstandard Part Approval Requests/Proposed Additions to Approval PPSL	DI-E-7028	--
3.3.2	Plan, Reliability Program	DI-R-7079	--
3.3.2.2	Repairable Items List	DI-V-7005	--
3.3.7	Requests, Nomenclature, and Identification Assignment and Approval	UDI-E-23090B	--
4.4.1	Report, First Article (Preproduction) Test	UDI-T-23790	--
4.5	Quality Inspection, Test Demonstration and Evaluation Report	DI-R-1725B	--
4.5.2.2	Reports, Reliability Test and Demonstration	DI-R-7034	--
4.5.4c	Plan, Corrective Action	DI-R-7038	--
4.6	Procedures, Test	UDI-T-21347	--
4.6	Procedures, Reliability Test	DI-R-7035	--

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, AMSDL. Copies of data items descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test approval, and disposition of the first article sample.

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6.4 Type designations. The type designation may be modified by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.7. The correct type number should be used on nameplates, shipping and instruction books, as applicable.

Preparing activity:

Navy - AS

(Project 5895-N312)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-85678(AS)	TRANSPONDER, RADAR TRACKING

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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